

**2022****North Sydney Boys High School****HSC Course Assessment Task 3-Trial Examination****MATHEMATICS EXTENSION 1****General instructions**

- Reading time - 10 minutes
- Working time - 2 hours.
- A reference sheet is provided
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- For questions in Section II, show relevant mathematical reasoning and/or calculations in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- At the conclusion of the examination, bundle any additional sheets used in the correct order within this paper and hand to examination supervisors.

Class (please tick)

- ☐ 12MM4.A2 – Mr Berry
- ☐ 12MM4.B2 – Ms Lee
- ☐ 12MM3.B2 – Mr Lin
- ☐ 12MM3.A2 – Mr Ireland
- ☐ 12MM3.C2 – Ms Moss
- ☐ 12MM4.C2 – Mr Umakanthan

**Student
Number**

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Marker's use only.

QUESTION	1-10	11	12	13	14	Total
MARKS	$\overline{10}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{70}$

Section I 10 Marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

1. A cubic function has only one distinct root. Which of the following could be true about the function.

- I** It's local maximum and local minimum values have opposite signs.
- II** It's local maximum and local minimum values have same signs.
- III** It has a horizontal point of inflexion.

- A. **I**
- B. **II only**
- C. **III only**
- D. **II or III**

2. If the parametric equations of a curve are $x = \sin t$ and $y = \cos^2 t + 1$, then the Cartesian equation of the curve is

- A. $y = x^2 - 1$
- B. $y = 1 - x^2$
- C. $y = 2 - x^2$
- D. $y = x^2 - 1$

3. Which of the set of 3 numbers could be the roots of the polynomial equation $x^3 + ax^2 - 41x + 42 = 0$?

- A. 2, 3, 7
- B. 1, -6, 7
- C. -1, -2, 21
- D. -1, -3, -14.

4. What is the value of $\int_0^{\frac{\sqrt{3}}{2}} \sqrt{1-y^2} dy$
- A. $\frac{\pi}{6}$
- B. $\frac{\pi}{3}$
- C. $\frac{\pi}{6} + \frac{\sqrt{3}}{8}$
- D. $\frac{\pi}{3} + \frac{\sqrt{3}}{8}$
5. When a polynomial $(3x^2 + 8x - 3)$ is multiplied by $(px - 1)$ and the resulting product is divided by $(x + 1)$ the remainder is 24. What is the value of p ?
- A. -4
- B. 2
- C. 4
- D. $\frac{11}{4}$
6. The largest value obtained by $3 \cos^2 x + 2 \sin x + 1$ equals
- A. $\frac{11}{5}$
- B. $\frac{13}{3}$
- C. $\frac{12}{5}$
- D. $\frac{14}{9}$
7. In the range $0 \leq x \leq 2\pi$, the equation $2^{\sin^2 x} + 2^{\cos^2 x} = 2$
- A. has no solutions
- B. has 1 solution
- C. has 2 solutions
- D. holds for all values of x

8. The graph of $f(x) = 0.6 \cos^{-1}(x - 1)$, defines a curve that, when rotated about the y-axis will produce a solid that is to be the shape and size of a new biscuit. Which integral expression will give the volume of the biscuit?

A. $\pi \int_0^{0.6} \left[\cos \left(\frac{5}{3}y \right) + 1 \right]^2 dy$

B. $\pi \int_0^{0.6} \left[\cos \left(\frac{3}{5}y \right) + 1 \right]^2 dy$

C. $\pi \int_0^{0.6\pi} \left[\cos \left(\frac{5}{3}y \right) + 1 \right]^2 dy$

D. $\pi \int_0^{0.6\pi} \left[\cos \left(\frac{3}{5}y \right) + 1 \right]^2 dy$

9. Let n be a positive integer. The coefficient of x^3y^5 in the expansion of $(1 + xy + y^2)^n$ equals

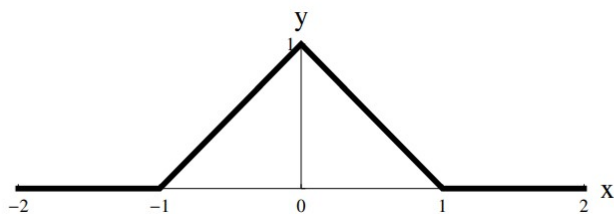
A. n

B. 2^n

C. $\binom{n}{3} \binom{n}{5}$

D. $4 \binom{n}{4}$

10. The graph of the function $y = f(x)$ is sketched below. The value of $\int_{-1}^1 f(x^2 - 1)dx$ equals



A. 2

B. $\frac{1}{3}$

C. $\sqrt{2}$

D. $\frac{2}{3}$

Section II**60 Marks****Attempt Questions 11-14****Allow about 1 hour and 45 minutes for this section**

Answer each question in the appropriate writing booklets. Extra writing booklets are available.

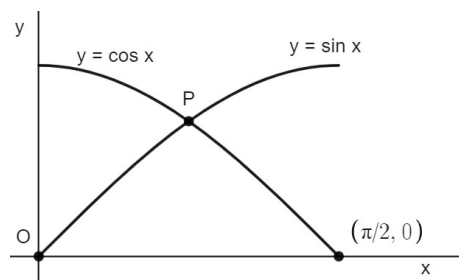
For questions in section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11. (15 Marks) Use the Question 11 Writing Booklet

- (a) (i) Solve $\frac{x+5}{x-1} \geq \frac{9}{x}$ for $x \neq 1, x \neq 0$ 3
- (ii) Hence find the set of values of x which satisfies $\frac{e^x+5}{e^x-1} \geq \frac{9}{e^x}$ 2
- (b) Find the remainder when $x^3 - 5x^2 + 7$ is divided by $(x-1)^2$ 3
- (c) (i) Express $f(\theta) = \sqrt{3} \sin 2\theta + \cos 2\theta$ in the form $P \cos(2\theta - \alpha)$, for $0^\circ \leq \alpha \leq 90^\circ$ 2
- (ii) Find the maximum and minimum values of $f(\theta)$ and the values of θ when they occur. 2
- (d) Evaluate $\cos\left(\tan^{-1} \frac{4}{3} - \cos^{-1} \frac{5}{13}\right)$. Exact answer required with working. 3

Question 12. (15 Marks) Use the Question 12 Writing Booklet.

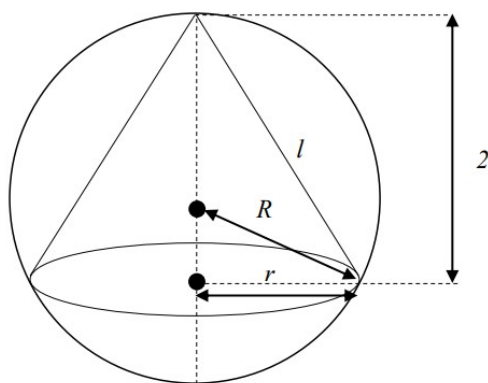
(a)



With origin O, the curves with equations $y = \sin x$ and $y = \cos x$ where $0 \leq x \leq \frac{1}{2}\pi$, meet at the point P with coordinates $\left(\frac{1}{4}\pi, \frac{1}{2}\sqrt{2}\right)$.

Find the exact value of the volume of the solid formed when the area bounded by the curves $y = \sin x$ and $y = \cos x$ and the x -axis is rotated about the x -axis by 2π radians. 4

(b) The diagram shows a right circular cone of height 2 units and radius r and slant height l inscribed in a sphere of radius R



(i) Show that $A = 4\pi\sqrt{R^2 - R}$, where A is the curved surface area of the cone. 2
(Curved surface area of a right circular cone of base radius r and slant height l is πrl)

(ii) If the volume of the sphere is increasing at the rate of $8 \text{ units}^3/\text{sec}$, 4
find the exact rate of change of A at the instant when $R = 2$ units.
(Volume of sphere is $\frac{4}{3}\pi R^3$)

(c) If $\operatorname{cosec} \theta - \cot \theta = \frac{4}{5}$, find the exact value of $\tan \theta$, without finding the value of θ 3

(d) Prove that $\sin^2 2\theta(\cot^2 \theta - \tan^2 \theta) = 4 \cos 2\theta$ 2

Question 13. (15 Marks) Use the Question 13 Writing Booklet.

(a) (i) Find $\int \frac{1 - \ln x}{x \ln x} dx$ 2

(ii) Use the substitution $u = x - 8$ to find $\int_8^{8.5} \frac{dx}{\sqrt{(7-x)(x-9)}}$ 3

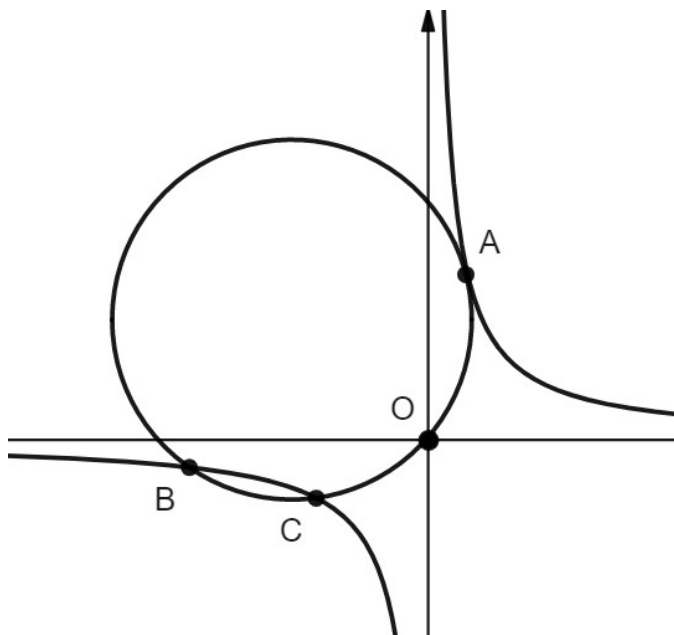
(b) α, β and γ are the roots of the equation $2x^3 + 5x^2 - 4x + 8 = 0$. Given that $\alpha^2 + \beta^2 + \gamma^2 = \frac{41}{4}$, 2
and $\alpha^3 + \beta^3 + \gamma^3 = \frac{-341}{8}$,

find the value of $\alpha^4 + \beta^4 + \gamma^4$

(c) Use vector methods to find the point P on the circle $(x - 5)^2 + (y - 4)^2 = 4$ which is closest to the 3
circle $(x - 1)^2 + (y - 1)^2 = 1$

You may assume that the point P and the centres of the two circles are collinear.

(d) A circle passing through the origin O is tangent to the hyperbola $xy = 1$ at A and intersects the 1
hyperbola again at two distinct points B and C . The co-ordinates of the points A, B and C are
 $(t, \frac{1}{t}), (t_1, \frac{1}{t_1})$ and $(t_2, \frac{1}{t_2})$ respectively



(i) Find the gradient of OA 1

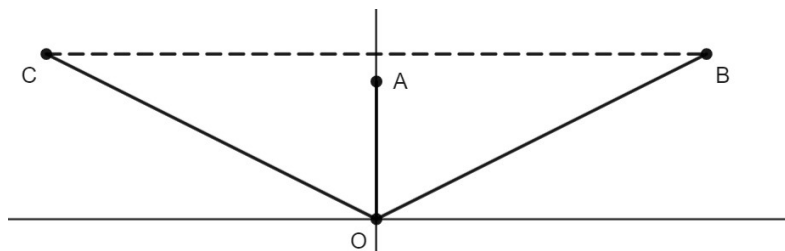
(ii) Show that the gradient of BC is $\frac{-1}{t_1 t_2}$ 1

(iii) Prove that OA is perpendicular to BC 3

Question 14. (15 Marks) Use the Question 14 Writing Booklet.

- (a) With reference to the origin O , the points A and B have position vectors $\underline{a}$ and $\underline{b}$ respectively, and O , A and B are non-collinear. The point C , with position vector $\underline{c}$, is the reflection of B in the line through O and A . 3

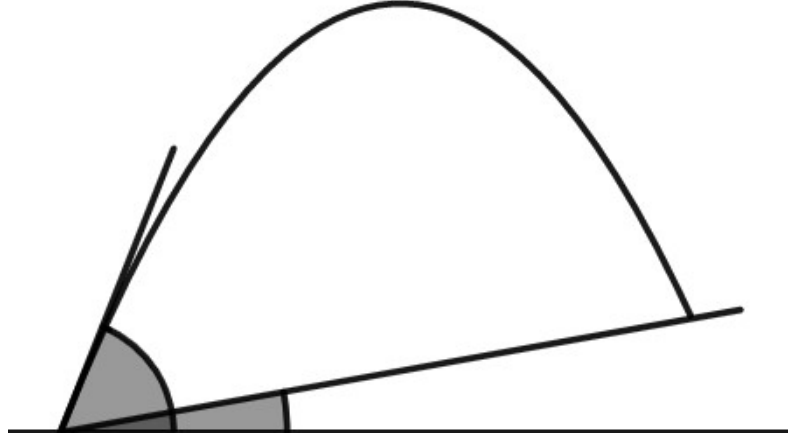
Show that $\underline{c}$ can be written in the form $\underline{c} = \lambda \underline{a} - \underline{b}$, where $\lambda = \frac{2\underline{a} \cdot \underline{b}}{\underline{a} \cdot \underline{a}}$



- (b) Prove by induction that for all positive integers n 3

$$n^2 - (n-1)^2 + (n-2)^2 - (n-3)^2 \cdots + (-1)^{n-1}(1)^2 = \frac{n}{2}(n+1)$$

- (c) Part of a golf course is on a hill which slopes at an angle α to the horizontal. The ball is hit straight up the hill with velocity $\underline{u} = [(|\underline{u}| \cos \theta)\underline{i} + (|\underline{u}| \sin \theta)\underline{j}] \text{ m/s}$. The position vector $\underline{r}$ of the particle at any time t is $\underline{r} = x\underline{i} + y\underline{j}$ and the acceleration due to gravity is $g \text{ ms}^{-2}$.



- (i) At any time t on its trajectory, derive the vector equation $\underline{r} = \underline{u}t - \frac{gt^2}{2}\underline{j}$ 3
- (ii) Hence show that $\frac{|\underline{r}|}{\cos \theta} = \frac{|\underline{u}|t}{\cos \alpha} = \frac{gt^2}{2 \sin(\theta - \alpha)}$ 2
- (iii) Hence show that the ball first lands at a distance of $\frac{2|\underline{u}|^2 \sin(\theta - \alpha) \cos \theta}{g \cos^2 \alpha}$ up the hill. 2
- (iv) For different values of θ show that the greatest distance up the hill the ball can be hit is $\frac{|\underline{u}|^2}{g(1 + \sin \alpha)}$. 2

End of Paper

Suggested solutions to 12 Maths Ext 1 NSWB 2022
Trial

1. D

2. C

3. B

4. ~~A~~ C

5. B

6. B

7. A

8. C

9. D

10. D.

11. (a) i)

$$\frac{x+5}{x-1} \geq \frac{9}{x} \quad x \neq 0, x \neq 1$$

Multiplying both sides by $x^2(x-1)^2$,

$$x^2(x+5)(x-1) \geq 9x(x-1)^2$$

$$\Rightarrow x(x-1)[9(x+1)-x(x+5)] \leq 0$$

$$\Rightarrow x(x-1)(-x^2+4x-9) \leq 0$$

$$\Rightarrow x(x-1)[(x-2)^2+5] \geq 0$$

$$\Rightarrow x(x-1) \geq 0$$

$$\Rightarrow \underline{x \leq 0} \text{ or } \underline{x > 1}$$

ii) $\frac{e^x+5}{e^x-1} \geq \frac{9}{e^x}$

$$\Rightarrow e^x < 0 \text{ or } e^x > 1$$

~~$\Rightarrow$ not true~~

$$e^x \neq 0 \therefore e^x > 1$$

$$\underline{x > 0}$$

(b) Let $P(x) = x^3 - 5x^2 + 7$

$$\Rightarrow P(x) = x^3 - 5x^2 + 7 = (x-1)^2 Q(x) + Ax + B$$

$$P(1) = 0 \Rightarrow A + B = 1 - 5 + 7 = 3 \quad \text{--- (1)}$$

$$P'(x) = 3x^2 - 10x = 2(x-1)Q(x) + (x-1)^2 Q'(x) + A$$

$$P'(1) = 0 \Rightarrow 3 - 10 = A \quad \text{--- (2)}$$

From (1) and (2)

$$A = -7 \text{ and } B = 10$$

$\therefore$ The remainder is $-7x + 10$

11) 11)

$$f(\theta) = \sqrt{3} \sin 2\theta + \cos 2\theta$$

$$= 2 \left[\frac{\sqrt{3}}{2} \sin 2\theta + \frac{1}{2} \cos 2\theta \right]$$

$$= 2 \left[\cos 2\theta \cos 60^\circ + \sin 2\theta \sin 60^\circ \right]$$

$$= 2 \cos(2\theta - 60^\circ)$$

$$= \cos(2\theta - \alpha) \text{ where } p=2 \text{ and } \alpha=60^\circ$$

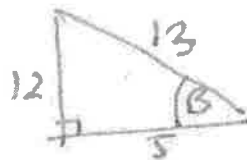
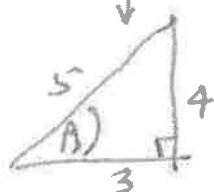
$$f(\theta) = 2 \cos(2\theta - 60^\circ)$$

$$f(\theta)_{\max} = 2 \text{ when } 2\theta - 60^\circ = 0 \Rightarrow \theta = 30^\circ$$

$$f(\theta)_{\min} = -2 \text{ when } 2\theta - 60^\circ = 180^\circ \Rightarrow \theta = 120^\circ$$

11) (d) Let $A = \tan^{-1}\left(\frac{4}{3}\right)$ and $B = \cos^{-1}\left(\frac{5}{13}\right)$

$$\Rightarrow \tan A = \frac{4}{3} \text{ and } \cos B = \frac{5}{13}$$



$$\begin{aligned} \cos(A-B) &= \cos A \cos B + \sin A \sin B \\ &= \frac{3}{5} \cdot \frac{5}{13} + \frac{4}{5} \times \frac{12}{13} \\ &= \frac{63}{65} \end{aligned}$$

$$(ii) \cos \left[\tan^{-1}\left(\frac{4}{3}\right) - \cos^{-1}\left(\frac{5}{13}\right) \right] = \frac{63}{65}$$

13. (a) Required volume = $V_1 + V_2$

$$V_1 = \pi \int_0^{\pi/4} \sin^2 x \, dx$$

$$= \frac{\pi}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi/4}$$

$$= \frac{\pi}{2} \left[\frac{\pi}{4} - \frac{1}{2} \right] \text{ units}^3.$$

$$V_2 = \pi \int_{\pi/4}^{\pi/2} \cos^2 x \, dx$$

$$= \frac{\pi}{2} \left[x + \frac{1}{2} \sin 2x \right]_{\pi/4}^{\pi/2}$$

$$= \frac{\pi}{2} \left[\frac{\pi}{2} - \left(\frac{\pi}{4} + \frac{1}{2} \right) \right]$$

$$= \frac{\pi}{2} \left[\frac{\pi}{4} - \frac{1}{2} \right]$$

$$\therefore V = V_1 + V_2$$

$$= \pi \left[\frac{\pi}{4} - \frac{1}{2} \right] \text{ units}^3$$

$$= \frac{\pi}{4} [\pi - 2] \text{ units}^3$$

(b) (i) By Pythagoras' Theorem $l = \sqrt{r^2 + 4}$

$$R^2 = r^2 + (2-r)^2 \Rightarrow r^2 = 4R - 4$$

$$\text{and } l = \sqrt{4R - 4 + 4} = \sqrt{4R}$$

$$A = \pi r l$$

$$= \pi \sqrt{4R - 4} \cdot \sqrt{4R}$$

$$= \underline{\underline{4\pi \sqrt{R^2 - R}}}$$

(b) (iii)

$$\begin{aligned}\frac{dA}{dt} &= \frac{dA}{dR} \times \frac{dR}{dV} \times \frac{dV}{dt} \\ &= \frac{2V(2R-1)}{\sqrt{R^2-1}} \cdot \frac{1}{4VK^2} \times 8\end{aligned}$$

When $R=2$

$$\begin{aligned}\frac{dA}{dt} &= \frac{2V(4-1)}{\sqrt{4-1}} \times \frac{1}{4V(4)} \times 8 \\ &= \frac{3\sqrt{2}}{2}\end{aligned}$$

$$(5) \quad \operatorname{cosec} \theta - \cot \theta = \frac{4}{5} \quad \text{--- (1)}$$

$$\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

$$\Rightarrow (\operatorname{cosec} \theta + \cot \theta)(\operatorname{cosec} \theta - \cot \theta) = 1$$

$$\Rightarrow (\operatorname{cosec} \theta + \cot \theta) \left(\frac{4}{5} \right) = 1$$

$$\Rightarrow \operatorname{cosec} \theta + \cot \theta = \frac{5}{4} \quad \text{--- (2)}$$

$$(2) - (1) \quad 2 \cot \theta = \frac{5}{4} - \frac{4}{5}$$

$$\cot \theta = \frac{9}{40}$$

$$\Rightarrow \tan \theta = \frac{40}{9}$$

$$(d) \quad \text{LHS} = \sin^2 2\theta (\cot^2 \theta - \tan^2 \theta)$$

$$= 4 \sin^4 \theta \cos^4 \theta \frac{(\cos^2 \theta - \sin^2 \theta)}{\sin^4 \theta \cos^4 \theta}$$

$$= 4 (\cos^2 \theta + \sin^2 \theta)(\cos^2 \theta - \sin^2 \theta)$$

$$= 4 \cos 2\theta$$

$$= \text{RHS}$$

Q13

(i) $\int \frac{1 - \ln x}{x \ln x} dx$

$$= \int \frac{1}{x \ln x} - \frac{1}{x} dx$$

$$= \ln(-\ln x) - \ln x + C$$

$$= \ln\left(\frac{\ln x}{x}\right) + C$$

(ii) $\int_8^{8.5} \frac{dx}{\sqrt{(7-x)(x-6)}}$

Let $u = x - 8$

$$\frac{du}{dx} = 1$$

$x = 8, u = 0$

$x = 8.5, u = 0.5$

$$= \int_0^{0.5} \frac{du}{\sqrt{-(u+1)(u-1)}}$$

$$= \int_0^{0.5} \frac{du}{\sqrt{1-u^2}}$$

$$= \left[\sin^{-1}(u) \right]_0^{0.5}$$

$$= \frac{\pi}{6}$$

(iii) $2x^3 + 5x^2 - 4x + 8 = 0 \quad x, y, z$

$P(x) \neq 0 \Rightarrow x \neq 0$

$$\Rightarrow 2x^4 + 5x^3 - 4x^2 + 8x = 0$$

x is a root

$$\therefore 2x^4 + 5x^3 - 4x^2 + 8x = 0 \quad \text{--- (1)}$$

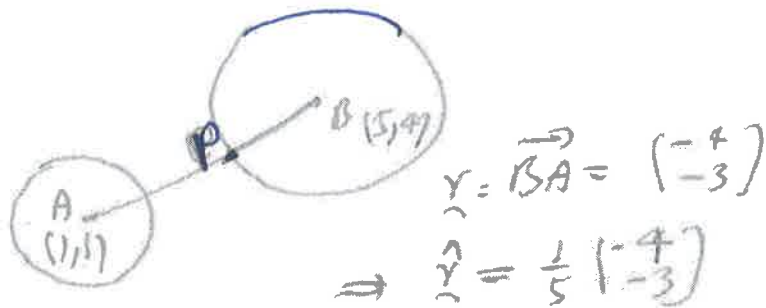
Similarly $2y^4 + 5y^3 - 4y^2 + 8y = 0 \quad \text{--- (2)}$

$2z^4 + 5z^3 - 4z^2 + 8z = 0 \quad \text{--- (3)}$

$$\textcircled{1} + \textcircled{2} + \textcircled{3} \quad 2(x^4 + y^4 + z^4) + 5\left(-\frac{341}{8}\right) - 4\left(\frac{41}{2}\right) + 8\left(-\frac{5}{2}\right) = 0$$

$$\Rightarrow 2(x^4 + y^4 + z^4) = 274 \frac{1}{8} \therefore x^4 + y^4 + z^4 = 137 \frac{1}{16}$$

13(c)

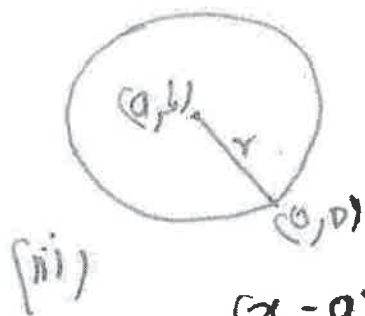


$$\vec{BC} = 2\hat{r} = \frac{2}{5} \begin{pmatrix} -4 \\ -3 \end{pmatrix}$$

$$\therefore \underline{P} = \begin{pmatrix} 5 \\ 4 \end{pmatrix} + \frac{2}{5} \begin{pmatrix} -4 \\ -3 \end{pmatrix} = \begin{pmatrix} 3.4 \\ 2.8 \end{pmatrix} \quad \therefore \underline{P(3.4, 2.8)}$$

(d) (i) gradient of OA = $m_{OA} = \frac{\frac{1}{t} - 0}{t - 0} = \frac{1}{t^2}$

(ii) $m_{BC} = \frac{\frac{1}{t_2} - \frac{1}{t_1}}{t_2 - t_1} = -\frac{1}{t_1 t_2}$



$$r = \sqrt{a^2 + b^2}$$

$\therefore$ The equation of the circle is

$$(x-a)^2 + (y-b)^2 = a^2 + b^2$$

$$\Rightarrow x^2 - 2ax + y^2 - 2by = 0$$

$A(t, \frac{1}{t})$ is on the circle

$$\Rightarrow t^2 - \frac{2a}{t} + \frac{1}{t^2} - \frac{2b}{t} = 0$$

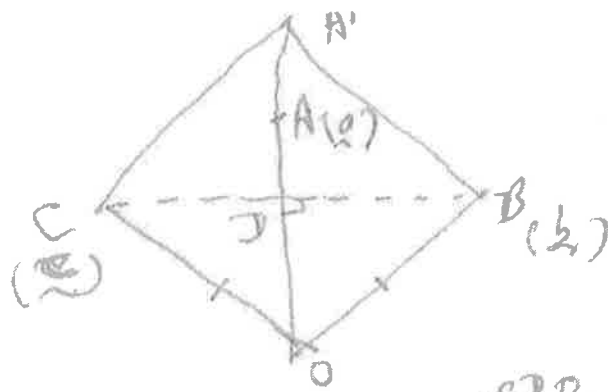
$$\Rightarrow t^4 - 2at^3 - 2bt + 1 = 0$$

Product of roots $t^2 t_1 t_2 = -1 \Rightarrow \frac{1}{t^2 t_1 t_2} = -1$

$$\Rightarrow \frac{1}{t^2} \times \frac{1}{t_1 t_2} = -1$$

$$\Rightarrow m_{OA} \times m_{BC} = -1 \Rightarrow \underline{OA \perp BC}$$

Q14. (a)



$$OB = OC, \angle CO = \angle BO = 90^\circ$$

$\Rightarrow$ ~~OA~~ OA is the bisector of $\angle BOC$

So that A is on the diagonal OA' of the parallelogram OBA'C

(I) is a rhombus, $OB = OC$

$$\therefore b + c = \lambda a$$

$$BC \perp OA \Rightarrow (b - c) \cdot a = 0$$

$$\Rightarrow [a - a] \cdot a = 0 \checkmark$$

$$\Rightarrow [b - (\lambda a - b)] \cdot a = 0$$

$$\Rightarrow [2b - \lambda a] \cdot a = 0 \checkmark$$

$$\Rightarrow \lambda = \frac{2b \cdot a}{a \cdot a} //$$

Q14 (b) Let P_n be the statement
 $n^2 - (n-1)^2 + (n-2)^2 - (n-3)^2 + \dots + (-1)^{n-1} (1)^2 = \frac{n}{2}(n+1)$ $n \in \mathbb{Z}^+$

When $n=1$,

$$\text{LHS} = (-1)^0 (1)^2 = 1, \quad \text{RHS} = \frac{1}{2}(1+1) = 1 \Rightarrow \text{LHS} = \text{RHS}$$

$\therefore P_1$ is true.

Assuming P_k is true for some $k \in \mathbb{Z}^+$

$$\text{i.e. } k^2 - (k-1)^2 + (k-2)^2 - \dots + (-1)^{k-2} (1)^2 = \frac{k}{2}(k+1)$$

To prove that P_{k+1} is true,

$$(k+1)^2 - k^2 + (k-1)^2 - (k-2)^2 - \dots + (-1)^{(k+1)-1} (1)^2 = \frac{(k+1)}{2}(k+2)$$

$$\text{i.e. } (k+1)^2 - k^2 + (k-1)^2 - \dots + (-1)^k (-1)^2 = \frac{(k+1)}{2}(k+2)$$

$$\text{LHS} = (k+1)^2 - k^2 + (k-1)^2 - \dots + (-1)^k (-1)^2$$

$$= \frac{(k+1)}{2} [2(k+1) - k]$$

$$= \frac{(k+1)}{2} [k+2]$$

$$= \text{RHS}$$

$\therefore P_k$ is true $\Rightarrow P_{k+1}$ is true

Since P_1 is true, by the method of mathematical induction,
 P_n is true for all $n \in \mathbb{Z}^+$.

14(c)

$$\underline{r} = x\underline{i} + y\underline{j}$$

$$\Rightarrow \underline{\dot{r}} = \dot{x}\underline{i} + \dot{y}\underline{j}$$

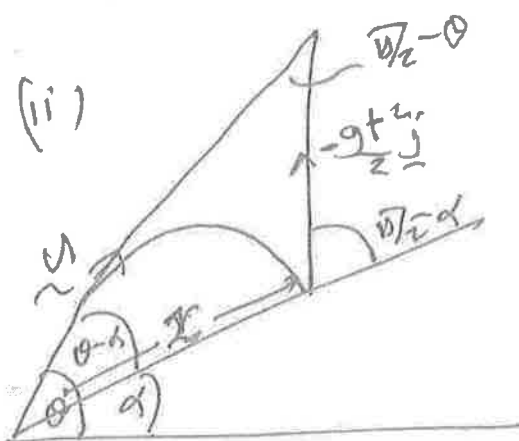
$$= 0\underline{i} - g\underline{j}$$

$$\Rightarrow \underline{\dot{r}} = -gt\underline{j} + c$$

$$t=0, \underline{\dot{r}} = u \Rightarrow c = \underline{u}$$

$$\underline{\dot{r}} = -gt\underline{j} + \underline{u}$$

$$\Rightarrow \underline{r} = -\frac{gt^2}{2}\underline{j} + \underline{u} \cdot t$$



Applying Sine Rule in $\triangle ABC$,

$$\frac{|\underline{r}|}{\sin(\frac{\pi}{2} - \theta)} = \frac{|\underline{u}|t}{\cos \alpha} = \frac{\frac{1}{2}gt^2}{\sin(\theta - \alpha)}$$

$$t = \frac{2|\underline{u}| \sin(\theta - \alpha)}{g \cos \alpha}$$

(iii)

$$\Rightarrow \frac{|\underline{r}|}{\cos \theta} = \frac{|\underline{u}|}{\cos \alpha} \frac{2|\underline{u}| \sin(\theta - \alpha)}{g \cos \alpha}$$

$$\Rightarrow |\underline{r}| = \frac{2|\underline{u}|^2 \sin(\theta - \alpha) \cos \theta}{g \cos^2 \alpha}$$

$$(iv) \quad r = \frac{2u^2 \sin(\theta - \alpha) \cos \theta}{g \cos^2 \alpha}$$

$$= \frac{u^2}{g \cos^2 \alpha} [\sin(2\theta - \alpha) \neq \sin \alpha]$$

$$= \frac{u^2}{g \cos^2 \alpha} [1 - \sin \alpha]$$

$$= \frac{u^2 (1 - \sin \alpha) (1 + \sin \alpha)}{g \cos^2 \alpha (1 + \sin \alpha)}$$

$$= \frac{u^2}{g(1 + \sin \alpha)}$$

when $2\theta - \alpha = \frac{\pi}{2}$